

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA0704 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I LAKSHITHA SREE K(192511336)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Computer Networks - Short Answers

1. The file is divided into 104,858 transport segments, and the same number of frames are transmitted. The Data Link layer introduces a total overhead of 4,194,320 bytes (about 4 MB). The transmission time is approximately 25.2 seconds. The Transport layer provides reliable delivery using sequencing, acknowledgements, and retransmissions, while the Data Link layer ensures error detection and reliable frame delivery.
2. With a sliding window size of 6, transmitting 48 frames requires 8 transmission windows. If one frame is lost in every window, a total of 8 retransmissions are needed. Flow control prevents the sender from overwhelming the receiver, reduces congestion, and improves communication efficiency.
3. A 12,000-byte packet passing through an MTU of 1500 bytes is divided into 9 fragments. Since each fragment carries a 20-byte IP header, the total header overhead is 180 bytes. The Network layer performs fragmentation at the router and reassembles the fragments at the destination using fragment offsets and identification fields.
4. The sender requires 8 transmission windows to send all 48 frames. One lost frame in each window results in 8 retransmissions. Flow control ensures that data is transmitted at a manageable rate, avoiding buffer overflow and improving reliable communication.
5. Synchronization checkpoints are created after every 60 MB, so 6 checkpoints are completed before the failure at 390 MB. Only the last 30 MB must be retransmitted after recovery. The Session layer uses synchronization points to resume communication without restarting the entire transfer.
6. Compressing a 900 MB file by 40% reduces it to 540 MB. Encryption adds 5% overhead, making the encrypted size 567 MB. The overall reduction from the original size is 37%. The Presentation layer improves efficiency through compression and ensures security using encryption.
7. A 3 GB file divided into 3 MB requests generates 1,024 requests. At an effective throughput of 120 Mbps, the estimated transmission time is about 204.8 seconds. The Application layer provides services such as file transfer, user authentication, and reliable communication through FTP.
8. The total end-to-end delay is 45 ms, including processing, transmission, and propagation delays. Each OSI layer performs specific functions such as formatting data, routing packets, and framing information. Together they ensure accurate and efficient communication.

9. A total of 52,429 frames are required to transfer the data. The protocol overhead transmitted is 4,194,320 bytes, and the transmission efficiency is approximately 95.24%. Although protocol overhead consumes bandwidth, it is essential for addressing, error detection, and reliable communication.

10. After 30% compression, the 240 MB file becomes 168 MB. The compressed data is divided into 125,830 segments, with a protocol overhead of 7,549,800 bytes. The total data transmitted is approximately 175.55 MB. The OSI layers work together to provide compression, segmentation, framing, secure transmission, and reliable delivery.